

SMART CAMPUS COMPANION

Design Document

Multi-Platform Software Development for Real-World Problems (CSE)

Problem 1: Smart Campus Companion — ASTU

Date: May 21, 2026

Platform: Flutter (Mobile) · Firebase (Backend) · OpenStreetMap

1. Introduction

This Design Document covers all design decisions made for the Smart Campus Companion mobile application. It includes the system architecture, database schema (ERD), UI/UX wireframe descriptions, and full API specification. All design decisions are grounded in the findings from the Analysis & Requirements Document.

Component	Technology Decision
Mobile platform	Flutter (primary) — Android & iOS
Backend	Firebase Firestore + Firebase Cloud Messaging
Maps	Google Maps SDK (google_maps_flutter package)
Authentication	JWT-based, role-separated (Student / Staff)
Offline support	Flutter Hive for local caching (innovation feature)
Version control	Git — one branch per feature

2. System Architecture

The application follows a three-layer architecture: Client, Backend, and Data. Each layer has a single responsibility and communicates through well-defined interfaces.

2.1 Architecture Layers

Layer	Responsibility & Components
Client layer	Flutter mobile app (students) + Staff admin panel (staff) + Optional web app (admin portal). All communicate with the backend via HTTPS REST API with JWT authentication.
Backend layer	Three services behind an API Gateway: Auth Service (login, JWT, role-based access), Data Service (schedules, announcements, directory, map), and Notification Service (FCM push delivery, retry on failure).
Data layer	PostgreSQL or Firebase Firestore for structured data. Firebase Storage for map assets and media. Firebase Cloud Messaging (FCM) for push notification delivery to Android and iOS devices.

2.2 Key Architectural Decisions

- API Gateway — all client requests pass through a single gateway for rate limiting, HTTPS enforcement, and JWT validation before reaching any service.
- Role-based access control — enforced at the API level. A student JWT cannot call staff-only endpoints (403 Forbidden). This is separate from UI-level hiding.
- Offline-first (Innovation Feature) — the Flutter app caches the last-synced timetable and announcements using Hive local storage. Students see their schedule even with no internet connection.
- FCM for push notifications — Firebase Cloud Messaging delivers alerts within 30 seconds of a staff post (NFR-02). Device tokens are stored per user in the DEVICE_TOKEN table.
- Google Maps integration — campus map uses the google_maps_flutter package locked to ASTU coordinates (8.5644N, 39.2921E) at zoom level 17–18. CameraTargetBounds restricts panning outside campus.

3. Database Schema (ERD)

The database contains seven tables. All primary keys are UUIDs. All timestamps are stored in UTC. Foreign key relationships enforce referential integrity across the schema.

3.1 USER Table

Central table for both students and staff. The role field determines which features and API endpoints the user can access.

Column	Type & Description
id	UUID · Primary Key
full_name	VARCHAR — student or staff full name
email	VARCHAR UNIQUE — university email address
password_hash	VARCHAR — bcrypt hashed password
role	ENUM: student staff admin
department	VARCHAR — e.g. CSE, CIVIL, CHEM
year_of_study	VARCHAR — null for staff
phone_number	VARCHAR — optional
avatar_url	VARCHAR — Firebase Storage URL
created_at / updated_at	TIMESTAMP WITH TIME ZONE

3.2 SCHEDULE Table

Stores each student's individual class slots. The status field is updated by the backend when a staff member posts a room change or cancellation announcement.

Column	Type & Description
id	UUID · Primary Key
user_id	UUID · FK → USER.id
course_name	VARCHAR — e.g. Data Structures
course_code	VARCHAR — e.g. CSE-302
room	VARCHAR — e.g. Room 204
building	VARCHAR — e.g. Block B
day_of_week	ENUM: monday tuesday ... sunday
start_time / end_time	TIME — 24hr format
lecturer_name	VARCHAR
status	ENUM: active cancelled room_changed
is_active	BOOLEAN — false for removed courses
created_at	TIMESTAMP WITH TIME ZONE

3.3 ANNOUNCEMENT Table

Stores all staff-posted announcements. The target_department field allows department-specific or university-wide broadcasting. is_urgent triggers immediate FCM push to all registered devices.

Column	Type & Description
id	UUID · Primary Key
posted_by	UUID · FK → USER.id (staff only)
title	VARCHAR — announcement headline
body	TEXT — full announcement content
category	ENUM: urgent academic general exam
target_department	VARCHAR — department code or ALL
is_urgent	BOOLEAN — triggers immediate push notification
published_at	TIMESTAMP WITH TIME ZONE
expires_at	TIMESTAMP — null for no expiry

created_at	TIMESTAMP WITH TIME ZONE
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3.4 NOTIFICATION Table

One row per user per announcement. This is the delivery log that powers the notification history screen (FR-10) and the analytics read-rate calculation for staff.

Column	Type & Description
id	UUID · Primary Key
user_id	UUID · FK → USER.id
announcement_id	UUID · FK → ANNOUNCEMENT.id
title	VARCHAR — denormalized for quick display
message	VARCHAR — short preview text
is_read	BOOLEAN — updated when student opens notification
sent_at	TIMESTAMP WITH TIME ZONE
read_at	TIMESTAMP — null until opened

3.5 CAMPUS_LOCATION Table

Stores every building, office, lab, and room on the ASTU campus with GPS coordinates for Google Maps marker placement.

Column	Type & Description
id	UUID · Primary Key
name	VARCHAR — e.g. Room 204, Registrar Office
building_code	VARCHAR — e.g. BLOCK-B, ADMIN, MAIN
location_type	ENUM: building office lab lecture_hall service
floor	VARCHAR — e.g. 2nd, Ground
latitude / longitude	FLOAT — GPS coordinates (ASTU campus)
description	TEXT — directions and notes
updated_at	TIMESTAMP WITH TIME ZONE

3.6 STAFF_DIRECTORY Table

Extends the USER table with office-specific information for staff members. Linked one-to-one with USER via user_id.

Column	Type & Description
id	UUID · Primary Key
user_id	UUID · FK → USER.id (one-to-one)
office_location	VARCHAR — e.g. Block B · Room 312 · 3rd floor
office_hours	VARCHAR — e.g. Mon & Wed 2:00–4:00 PM
phone_extension	VARCHAR — campus extension number
personal_email	VARCHAR — optional direct email
specialization	VARCHAR — research/teaching areas
updated_at	TIMESTAMP WITH TIME ZONE

3.7 DEVICE_TOKEN Table

Stores FCM device tokens for push notification delivery. One user may have multiple tokens (different devices). Tokens are refreshed on each login.

Column	Type & Description
id	UUID · Primary Key
user_id	UUID · FK → USER.id
fcm_token	VARCHAR — Firebase Cloud Messaging token
platform	ENUM: android ios
registered_at	TIMESTAMP WITH TIME ZONE
last_active	TIMESTAMP — updated on each app open

3.8 Table Relationships

Relationship	Cardinality & Description
USER → SCHEDULE	One-to-many — one student has many class slots
USER → NOTIFICATION	One-to-many — one user receives many notifications
USER → STAFF_DIRECTORY	One-to-one — each staff member has one directory profile
USER → DEVICE_TOKEN	One-to-many — one user can register multiple devices
USER → ANNOUNCEMENT	One-to-many — one staff member can post many announcements
ANNOUNCEMENT → NOTIFICATION	One-to-many — one announcement triggers many notifications

4. UI/UX Wireframes

Seven screen groups were designed covering the complete user journey for both students and staff. All screens follow a consistent design language: Teal (#1D9E75) as the primary color, green left-border for active/current items, red for urgent alerts, amber for warnings, and a consistent bottom navigation bar across all student screens.

4.1 Login & Onboarding (5 screens)

Screen	Description
Screen 1 — Role selector	Splash screen with Student / Staff toggle. Button label updates dynamically. Routes to respective login flow.
Screen 2 — Login form	Email/password fields with university SSO option. Shows error state for wrong credentials. Forgot password link included.
Screen 3 — Onboarding slide 1	Never miss a thing — introduces push notifications. Skippable. Progress dots shown.
Screen 4 — Onboarding slide 2	Your schedule, always with you — introduces timetable and offline access.
Screen 5 — Onboarding slide 3	Find anything on campus — introduces map and staff directory.

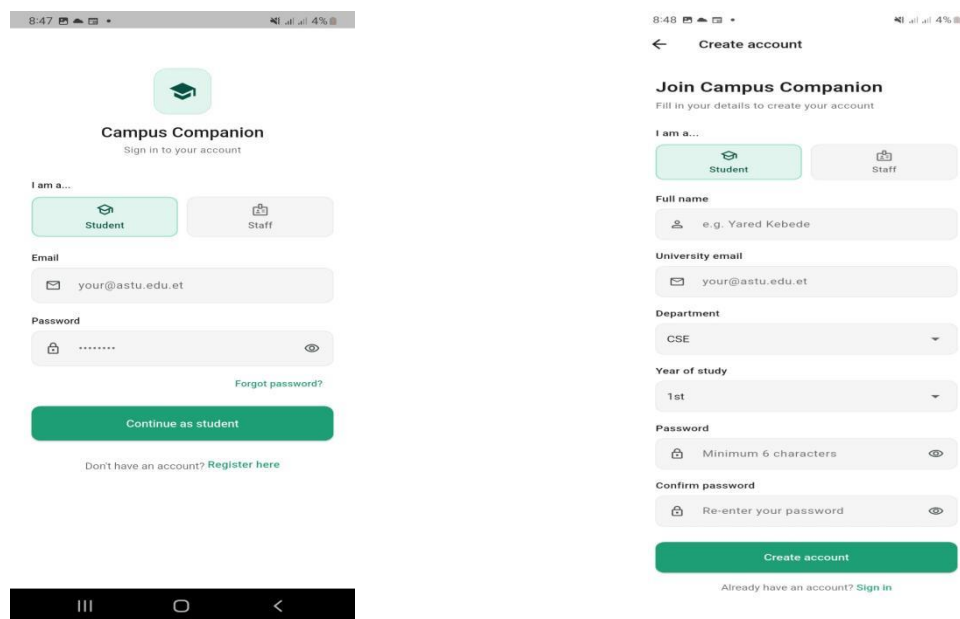


Figure 4.1 — Login screen (left) and Create Account screen (right)

4.2 Student Dashboard (3 states)

State	Description
Normal state	Greeting with name and day summary. Three stat cards (classes today, new alerts, days to exam). Today's schedule with color-coded class cards (green = now, amber = next). Recent announcements

	preview. Bottom navigation bar.
Urgent alert state	Red banner at top for room changes. Amber banner for quiz reminders. Bell icon shows red dot badge. Affected class card updates room detail in red with arrow indicating change. This directly solves Pain Points 1, 2, and 3 from research.
Offline state	Green offline banner at top. Schedule shows from cache. Announcements show graceful empty state explaining offline limitation. Satisfies NFR-03 and the student survey request for offline functionality.

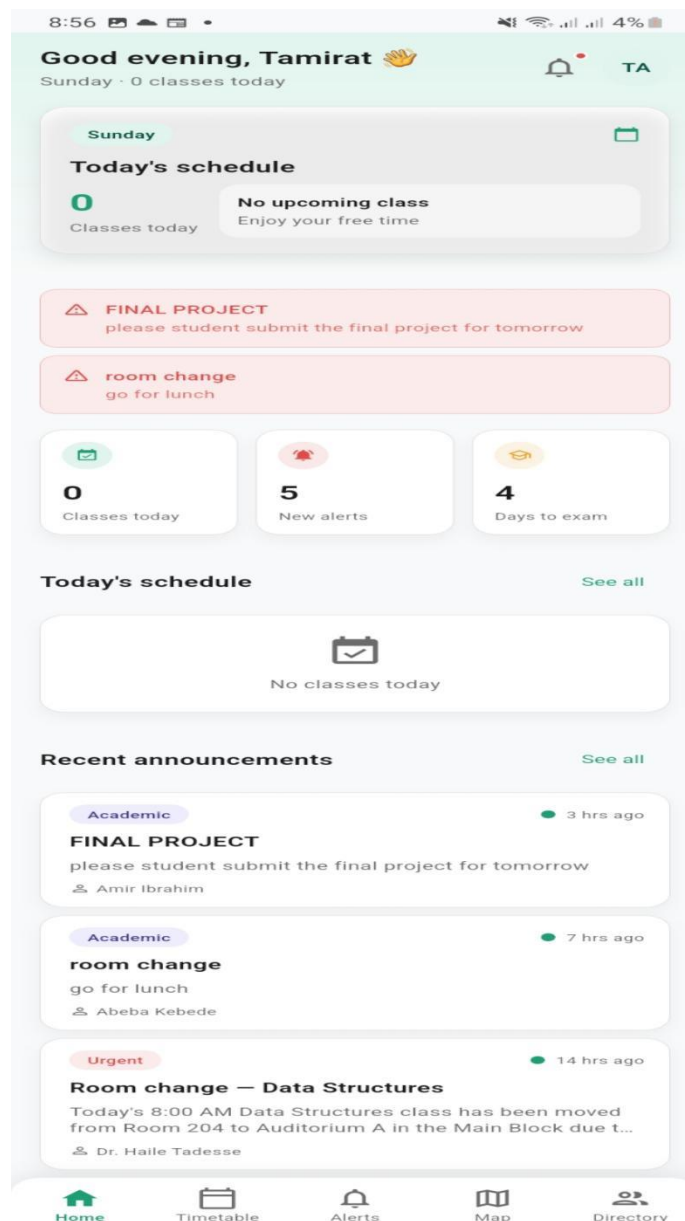


Figure 4.2 — Student Dashboard: today's schedule and recent announcements

4.3 Timetable & Academic Calendar (4 screens)

Screen	Description
Weekly view	Horizontal day strip at top. Current class has green left border. Three-tab navigation: Week / Calendar / Exams.
Calendar view	Full month grid. Green dots for class days, red dots for exam days. Tapping a date shows classes below the grid.
Exam schedule	All upcoming exams sorted by proximity with countdown (In 5 days, In 7 days). Red left border. Faded styling for distant exams.
Class detail view	Full class info with room, time, lecturer, course code. Two action

buttons: Find room on map (links to campus map) and View lecturer profile (links to directory).

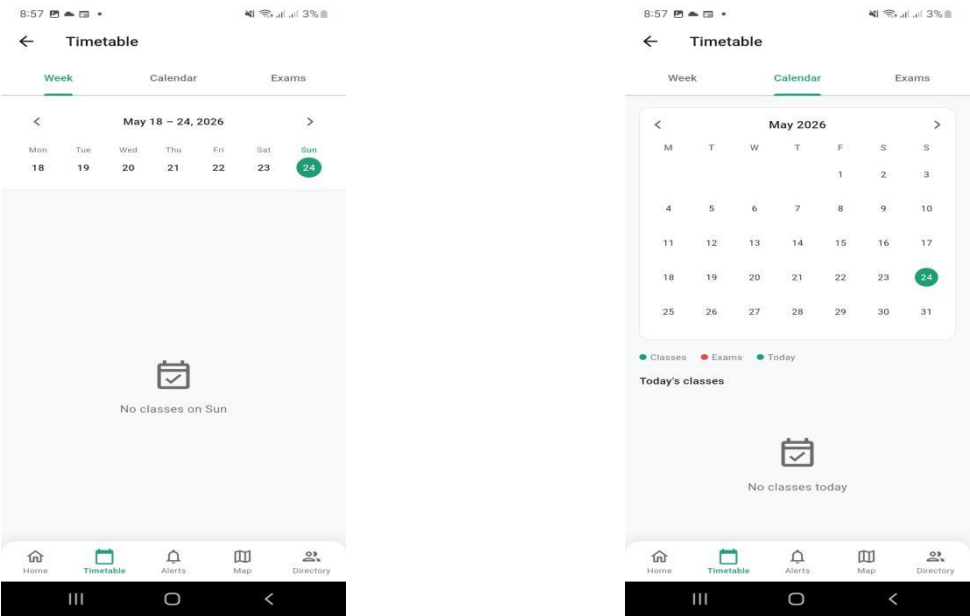


Figure 4.3 — Timetable Week view (left) and Calendar view (right)

4.4 Announcements Feed (4 screens)

Screen	Description
Feed	Search bar + filter chips (All, Urgent, Academic, Services, Exam). Unread items have green dot and shaded background. Urgent announcements pinned to top regardless of time.
Announcement detail	Full message text, poster name, timestamp. Find on map button for location-based announcements. Share button to forward official message to classmates (prevents garbled retelling).
Notification history	Grouped by day (Today, Yesterday, 2 days ago). Unread dot per item. Mark all read button. Satisfies FR-10.
Empty state	Friendly message when filter returns no results. Reinforces that urgent alerts will appear here immediately.

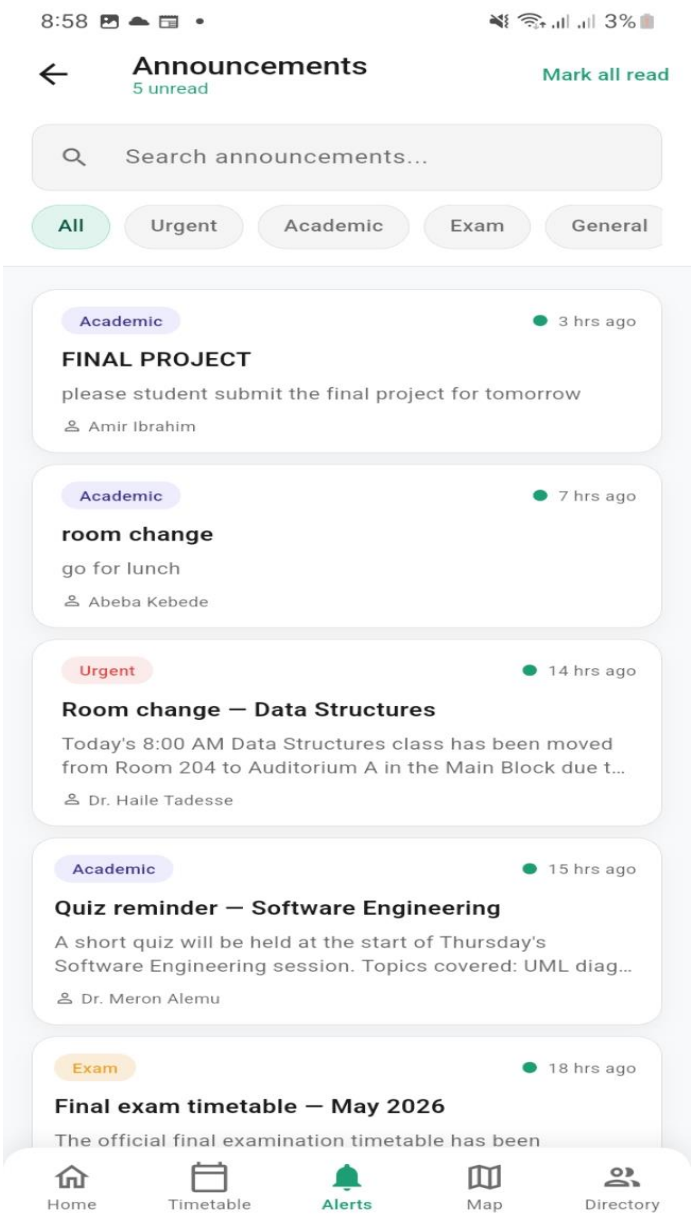


Figure 4.4 — Announcements feed with category filters and unread indicators

4.5 Campus Map (3 screens + real Google Maps)

Screen	Description
Default map view	Real Google Maps embedded via google_maps_flutter package. Locked to ASTU campus at 8.5644N, 39.2921E, zoom 17–18. Color-coded building markers. Search bar and filter chips overlaid at top. Zoom, pan, and current-location controls.
Location selected	Tapped marker highlights building. Bottom sheet slides up with room name, floor, capacity, walking directions, and estimated time. Two action buttons: Get directions and See today's classes here.
Search results	Mixed results list (buildings, people, rooms) with color-coded icons. Info

	bar explains to tap for map pin. Deep-links to selected location on the map.
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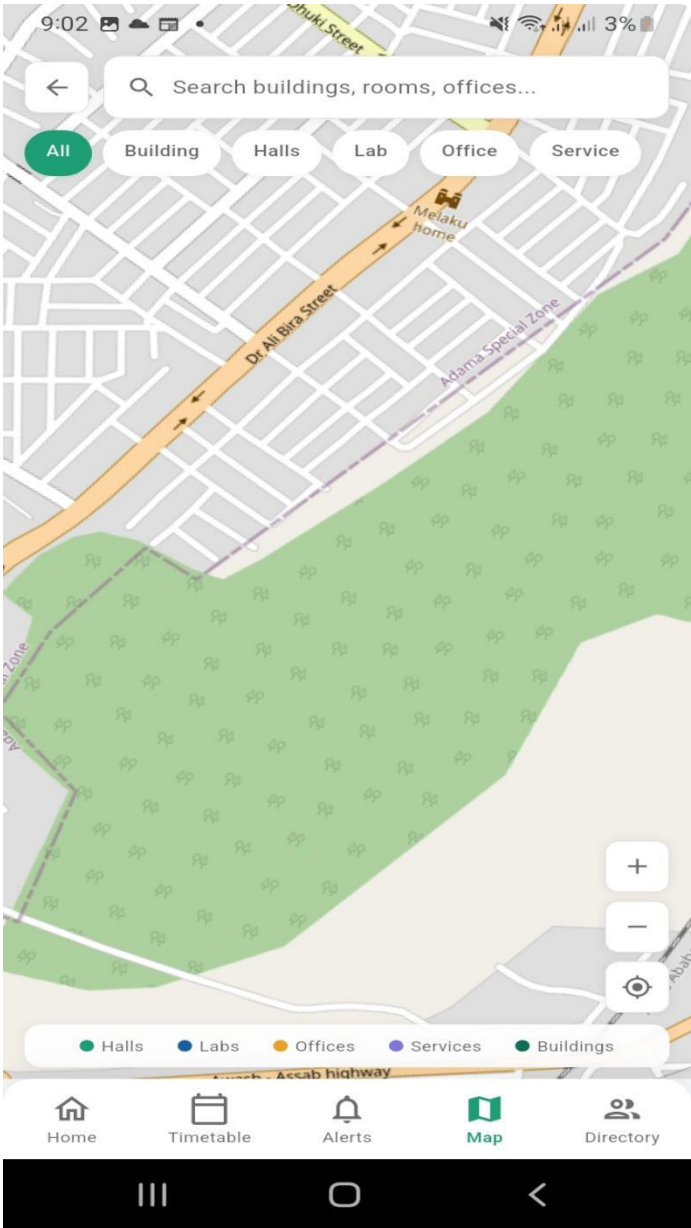


Figure 4.5 — Campus Map with building/lab/hall filter chips and color-coded legend

4.6 Staff Directory (3 screens)

Screen	Description
Main list	Search bar searching name, role, and department. Filter chips. Staff grouped by department. Initials avatar with color per department.

Staff profile	Contact card with email (tappable), office extension (tappable), exact office location, office hours, and courses taught. Find office on map and View this lecturer's classes buttons link to other features.
Department browser	Browse by school — Engineering & IT, Natural Sciences, Business, Humanities, Administrative. Shows staff count per department.

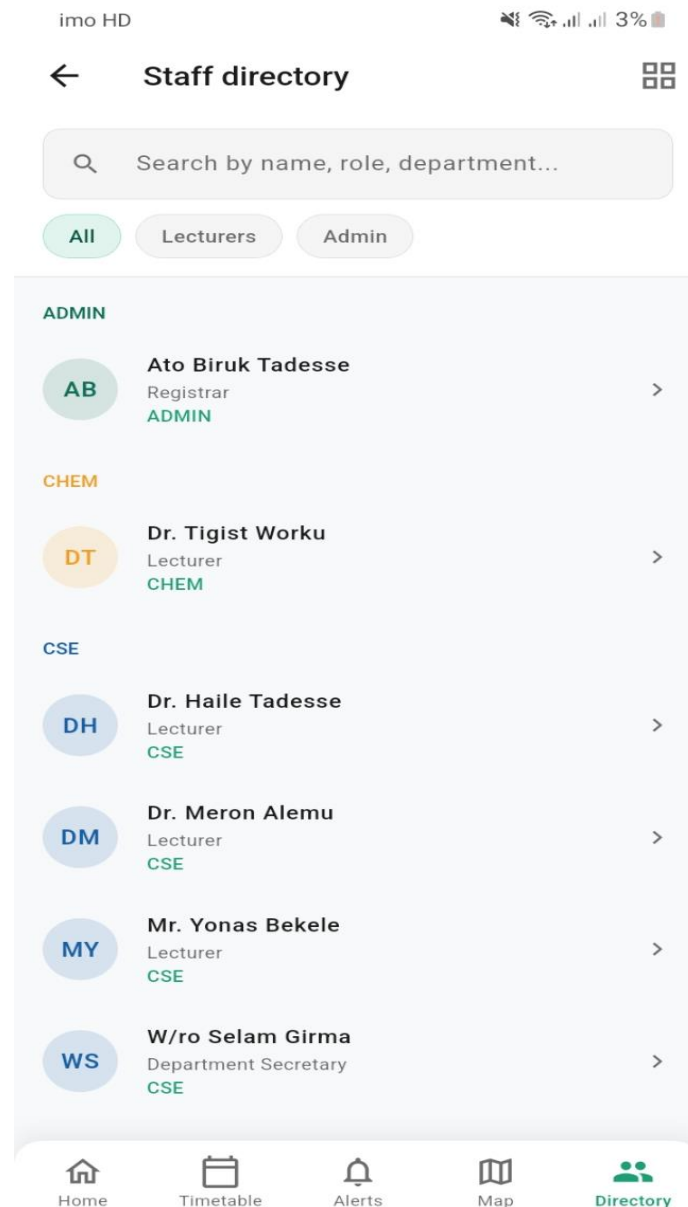


Figure 4.6 — Staff Directory grouped by department with search and filter

4.7 Staff Admin Panel (4 screens)

Screen	Description
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Staff dashboard	Stat cards: posts this week, students reached, read rate. Two prominent action buttons: Post announcement (green) and Post urgent alert (red). Recent posts list with edit/delete options and delivery stats.
Post announcement	Type selector (Urgent / Academic / General). Title and message fields. Send to field showing target audience and student count. Three toggles: send push notification, mark urgent, schedule for later. Single Send button. Designed for under 3 steps — satisfies NFR-04.
Sent confirmation	Success icon with student count delivered. Push notification confirmation. Back to dashboard and Post another buttons. Replaces the anxiety of the representative system.
Analytics	Summary stats: total posts, average read rate, total reached. Bar chart showing read rates by category (Urgent 94%, Academic 88%, General 76%). Top performing posts list with time-to-read metric.

5. API Design

The REST API follows standard HTTP conventions. All endpoints are prefixed with /v1/ for versioning. All requests and responses use JSON. All endpoints except /auth/login require a JWT Bearer token in the Authorization header.

5.1 API Overview

Property	Value
Base URL	https://api.campuscompanion.astu.edu.et/v1
Authentication	JWT Bearer token — expires after 24 hours (NFR-05)
Format	JSON · UTF-8 · HTTPS only
Versioning	/v1/ prefix — breaking changes get a new version number
Role enforcement	Student token on staff endpoint returns 403 Forbidden
Total endpoints	19 endpoints across 7 resource groups

5.2 Authentication Endpoints

Endpoint	Auth Level	Description
POST /auth/login	Public	Login with email + password + role. Returns JWT token and user profile.
POST /auth/logout	Student / Staff	Invalidates the current JWT token server-side.
POST /auth/register-device	Student / Staff	Registers an FCM device token for push notification delivery.

5.3 Schedule Endpoints

Endpoint	Auth Level	Description
GET /schedule/today	Student	Returns today's class list with room, time, lecturer, and status (active cancelled room_changed).
GET /schedule/week	Student	Returns full week timetable. Accepts ?date= query param for any week.
GET /schedule/exams	Student	Returns upcoming exams sorted by date with days_until countdown field.

5.4 Announcement Endpoints

Endpoint	Auth Level	Description
GET /announcements	Student	Returns paginated feed. Supports ?category= and ?department= filters. 20 per page.
GET /announcements/{id}	Student	Returns full announcement detail including poster info and read status.
POST /announcements	Staff	Creates and sends announcement. Returns students_notified count. Triggers FCM if send_push: true.
PUT /announcements/{id}	Staff	Updates title or body of an existing announcement.
DELETE /announcements/{id}	Staff	Permanently deletes an announcement.

5.5 Notification Endpoints

Endpoint	Auth Level	Description
GET /notifications	Student	Returns notification history with unread_count. Grouped by date on the client.
PUT /notifications/{id}/read	Student	Marks a single notification as read. Updates read_at timestamp.
PUT /notifications/read-all	Student	Marks all notifications as read in one call.

5.6 Campus Map Endpoints

Endpoint	Auth Level	Description
GET /locations	Student	Returns all campus locations with GPS coordinates. Supports ?type= and ?q= search filters.
GET /locations/{id}	Student	Returns single location detail including linked staff in that office.

5.7 Staff Directory Endpoints

Endpoint	Auth Level	Description
GET /staff	Student	Searches staff by name, role, or department. Supports ?q=, ?dept=, ?role= filters.
GET /staff/{id}	Student	Returns full staff profile including courses, office hours, and location_id for map linking.
PUT /staff/{id}	Staff	Allows staff to update their own office hours, location, and phone extension.

5.8 Analytics Endpoint

Endpoint	Auth Level	Description
GET /analytics/my-posts	Staff	Returns delivery stats for all staff's own announcements —

		total reach, read rates by category, and per-post performance.
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5.9 Standard Error Responses

HTTP Status	Meaning
400 Bad Request	Missing or invalid fields in the request body.
401 Unauthorized	Missing, expired, or malformed JWT token.
403 Forbidden	Valid token but insufficient role — e.g. student calling a staff endpoint.
404 Not Found	Requested resource does not exist.
500 Server Error	Unexpected failure — client should retry after a moment.

6. Design Summary

The following table maps each core feature to the design components that implement it, showing how the architecture, database, wireframes, and API all work together:

Feature	Design Components
Student dashboard with today's schedule	SCHEDULE table + GET /schedule/today + Dashboard wireframe (3 states)
Real-time push notifications	NOTIFICATION + DEVICE_TOKEN tables + Notification Service + FCM + POST /announcements + Staff admin wireframe
Announcements feed	ANNOUNCEMENT table + GET /announcements + Announcements wireframes (4 screens)
Campus map (real ASTU Google Maps)	CAMPUS_LOCATION table + GET /locations + google_maps_flutter package + Map wireframes
Staff & office directory	STAFF_DIRECTORY table + GET /staff + Directory wireframes (3 screens)
Offline timetable access (innovation)	Flutter Hive local cache + GET /schedule/week + Dashboard offline state wireframe
Staff admin dashboard	ANNOUNCEMENT table + POST /announcements + GET /analytics/my-posts + Staff admin wireframes (4 screens)

End of Design Document